

2. Lecture 4

 Bookmark this page

- Upgrade your course today
- 

 You are taking the free version of this course.
- 

 Upgrade today and, upon passing, receive your digital certificate signed by MIT faculty to highlight the knowledge and skills you’ve gained from this MITx course.

Upgrade

4-1

1/1 point (graded)
The image of the unit square (the set of x and y such that $0 \leq x \leq 1$ and $0 \leq y \leq 1$) under matrix multiplication by

$$\begin{pmatrix} 3 & -1 \\ -6 & 2 \end{pmatrix}$$

is a line segment. The rightmost endpoint of this segment is a point of the form $\begin{pmatrix} a \\ b \end{pmatrix}$. Find a .

$a =$ ✓ Answer: 3

Solution:

The image of $\begin{pmatrix} x \\ y \end{pmatrix}$ under this transformation is

$$\begin{pmatrix} 3 & -1 \\ -6 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3x - y \\ -6x + 2y \end{pmatrix}.$$

The first coordinate is maximized by letting $x = 1$ and $y = 0$. Hence $a = 3$.

You have used 1 of 4 attempts

Answers are displayed within the problem

4-2

5/5 points (graded)
True or false: If $\mathbf{A}$ is an $n \times n$ matrix, and $\mathbb{C}^n$ has a basis consisting of eigenvectors of $\mathbf{A}$, then $\mathbf{A}$ has n distinct eigenvalues.

☐ True

☒ False



Solution:

False.

If $\mathbf{A}$ has n distinct eigenvalues, then it is guaranteed that $\mathbb{C}^n$ has a basis consisting of eigenvectors. But there are also some matrices $\mathbf{A}$ with repeated eigenvalues (such as $\mathbf{I}$) such that $\mathbb{C}^n$ still has a basis consisting of eigenvectors.

You have used 1 of 1 attempt

Answers are displayed within the problem

4-3

5.0/5.0 points (graded)
What is the characteristic polynomial of $\begin{pmatrix} 1 & 4 \\ 5 & 9 \end{pmatrix}$?
(Type L to denote the variable λ in the polynomial.)

L^2-10*L-11

✔ Answer: L^2-10*L-11

L^2 - 10 \cdot L - 11

? FORMULA INPUT HELP

Solution:

The characteristic polynomial is $\lambda^2 - 10\lambda - 11$.

The characteristic polynomial of a 2×2 matrix $\mathbf{A}$ is

$$\lambda^2 - \operatorname{tr}(\mathbf{A}) \lambda + \det(\mathbf{A})$$

For the given matrix $\mathbf{A}$, the trace is 10 and the determinant is -11 .
Alternative solution: The characteristic polynomial is

$$\det(\lambda \mathbf{I} - \mathbf{A}) = \det \begin{pmatrix} \lambda - 1 & -4 \\ -5 & \lambda - 9 \end{pmatrix} = (\lambda - 1)(\lambda - 9) - (-4)(-5) = \lambda^2 - 10\lambda - 11.$$

(One could also use $\det(\mathbf{A} - \lambda \mathbf{I})$; there is no need to change the sign when the size of the matrix is even.)

Submit

You have used 1 of 7 attempts

Answers are displayed within the problem

4-4

5.0/5.0 points (graded)

What is the characteristic polynomial of $\begin{pmatrix} 0 & 438 & 691 \\ 0 & 1 & 300 \\ 0 & 0 & 2 \end{pmatrix}$?

(Type L to denote the variable λ in the polynomial.)

-L^3+3*L^2-2*L

✔ Answer: L^3-3*L^2+2*L

Solution:

The characteristic polynomial is $\lambda(\lambda - 1)(\lambda - 2) = \lambda^3 - 3\lambda^2 + 2\lambda$.

The matrix $\lambda \mathbf{I} - \mathbf{A}$ is upper triangular with diagonal entries λ , $\lambda - 1$, $\lambda - 2$, so its determinant is $\lambda(\lambda - 1)(\lambda - 2)$.

Submit

You have used 3 of 10 attempts

Answers are displayed within the problem

2. Lecture 4

Hide Discussion

Topic: Unit 2: Linear Algebra, Part 2 / 2. Lecture 4

Add a Post

Show all postsby recent activity

There are no posts in this topic yet.

◀ Previous

Next ▶

© All Rights Reserved

© MITx Online. All rights reserved except where noted.

[About Us](#)

[Terms of Service](#)

[Privacy Policy](#)

[Honor Code](#)

[Accessibility](#)

 Calculator